

14	D	Acceleration due to gravity experienced by the rock a_c = centripetal acceleration $= r\omega^2$ $= (3.40 \times 10^3 \times 10^3 + 1.72 \times 10^4 \times 10^3) \left(\frac{2\pi}{8.86 \times 10^4} \right)^2$ $= 0.1036 \text{ m s}^{-2}$ Since acceleration due to gravity is gravitational field strength, $a = g = \frac{GM}{r^2}$ $a \propto \frac{1}{r^2}$ $\frac{a'_c}{0.1036} = \left(\frac{3.40 \times 10^3 + 1.72 \times 10^4}{3.40 \times 10^3} \right)^2$ $a'_c = 3.80 \text{ m s}^{-2}$
15	D	In uniform circular motion, the satellite's velocity is constantly changing due